

Lazar Đoković

☎ +381 65 578 2007 | 🌐 Personal Website | ✉ Email | 🔗 LinkedIn | 🐙 GitHub

EDUCATION

Erasmus Mundus Joint Master in Artificial Intelligence (M.Sc.) Sep. 2024 – Aug. 2026
University of Ljubljana · Radboud University · Pompeu Fabra University · UCL · GPA: 9.75/10.0, cum laude

Master's thesis: A Memory-Based Deep Learning Method for Long-Term Visual Point Tracking

Supervisors: Prof. Matej Kristan and Dr. Alan Lukežič · Grade: 10/10

B.Sc. in Computer and Information Science Oct. 2021 – Aug. 2024
University of Ljubljana, Faculty of Computer and Information Science · GPA: 9.5/10.0 Ljubljana, Slovenia

Bachelor's thesis: Sarcasm Detection with Transfer Learning from Multiple Sources

Supervisor: Prof. Marko Robnik-Šikonja · Grade: 10/10

RESEARCH EXPERIENCE

Research Intern, VILAB (Summer@EPFL) Jul. 2026 – Present
EPFL | Supervisor: Prof. Amir Zamir Lausanne, Switzerland

- Analyzing FlexTok's variable-length discrete latent space through **random-sequence decoding and token-swapping**, showing that arbitrary sequences often produce implausible images while individual tokens can transfer localized visual content.
- Evaluated all released FlexTok checkpoints on COCO across token budgets using FID; currently **training hierarchical tokenization models** to study how effectively FlexTok representations can be distilled into fewer tokens.
- Working toward training FlexTok across multiple modalities and distilling the resulting representations into a **unified multimodal representation** for downstream tasks.

Dense Memory-Based Visual Point Tracking Oct. 2025 – Jul. 2026
ViCoS, University of Ljubljana | Supervisors: Prof. Matej Kristan & Dr. Alan Lukežič Ljubljana, Slovenia

- Developed an attention-based global matching stage for robust long-range point correspondence and recovery after extended occlusions, replacing the localized refinement used in previous dense trackers.
- Introduced a **coarse-to-fine appearance memory** with visibility- and confidence-gated updates, adapting to appearance changes while filtering unreliable and background-contaminated features.
- Proposed **DeMeTer**, a dense memory-based point tracker that **establishes a new state of the art** on three of four *TAP-Vid/RoboTAP* benchmarks, delivering a 4.1% relative improvement in δ_{avg} over the strongest prior online method while operating in real time; manuscript in preparation.

XFeat Revisited: Reproducibility and Robustness Evaluation Oct. 2025 – Feb. 2026
Data Science Lab, University of Ljubljana Ljubljana, Slovenia

- Re-implemented XFeat and reproduced relative-pose, homography, and visual-localization evaluations on *MegaDepth-1500*, *ScanNet-1500*, *HPatches*, and *Aachen*.
- Extended the study with **novel architectural ablations and zero-shot out-of-distribution evaluations**, showing that the keypoint branch primarily benefits semi-dense matching and that performance degrades under severe viewpoint and modality shifts.
- Published the study in **TMLR** with a Reproducibility Certification; accepted for presentation at **NeurIPS 2026**, MLRC Reproducibility Track.

Sarcasm Detection in a Less-Resourced Language Oct. 2023 – Aug. 2024
ML and NLP Lab, University of Ljubljana | Supervisor: Prof. Marko Robnik-Šikonja Ljubljana, Slovenia

- Created a Slovene version of *iSarcasmEval* using neural machine translation and evaluated cross-lingual transfer across monolingual and multilingual Transformer models.
- Fine-tuned models across multiple scales and pretraining strategies and developed an ensemble that achieved an F_1 score of 0.765, **approaching annotator agreement** in the source language; published and presented the work at **SCAI 2024**.

PUBLICATIONS

[1] **L. Đoković** and A. Lin. *XFeat Revisited: Reproducibility and Evaluation of a Lightweight Image Matcher*. Transactions on Machine Learning Research (**TMLR**), 2026. Reproducibility Certification. Selected for presentation at the **MLRC Reproducibility Track, NeurIPS 2026**.

[2] **L. Đoković** and M. Robnik-Šikonja. *Sarcasm Detection in a Less-Resourced Language*. Proceedings of the Slovenian Conference on Artificial Intelligence (**SCAI**), pp. 19–22, 2024.

INDUSTRY EXPERIENCE

Microsoft Data Scientist Intern <i>Supervisor: Dr. Maja Stikić</i>	Jun. 2025 – Sep. 2025 <i>Belgrade, Serbia</i>
• Worked on Code Copilot, an agentic system that converts large enterprise codebases into code graphs to support software understanding and modernization. • Developed an agent that traverses code graphs sequentially by following relationships, replacing fixed Cypher retrieval that returned excessive irrelevant context. • Evaluated the approach with subject-matter experts and LLM-as-a-judge pipelines, improving user satisfaction while using approximately 11% of the context required by the previous method; contributed to its deployment to thousands of engineers.	
Teads AI Scientist Intern <i>Supervisor: Prof. Jure Demšar</i>	Feb. 2025 – Jun. 2025 <i>Remote, Slovenia</i>
• Studied click-through-rate scaling laws for Teads’ production Deep & Cross prediction model in an online-learning setting. • Ran large-scale experiments on 13.5 million anonymized time-series ad-interaction samples to identify batch-size and learning-rate hyperparameters that improved training speed and convergence.	
University of Ljubljana Data Curation Intern <i>Graphics and Multimedia Lab Supervisor: Prof. Matija Marolt</i>	Jun. 2022 – Aug. 2022 <i>Ljubljana, Slovenia</i>
• Digitized 20th-century Slovene music magazines and Yugoslav parliamentary records into searchable, linguistically annotated archives, preserving historically valuable material for research.	

AWARDS & HONORS

Summer@EPFL Scholarship EPFL	Jul. 2026
• Selected for EPFL’s competitive summer research program (acceptance rate of 1.2%).	
Best Poster Award Erasmus Mundus Joint Master in Artificial Intelligence	Apr. 2026
• Awarded for presenting research on dense memory-based visual point tracking (1st / 25).	
1st Place, Programmed Art Competition University College London and G-Research	Jul. 2025
• Created a visualization of language-model bias using multidimensional random walks (1st / 30).	
Full Erasmus Mundus Scholarship European Commission	Sep. 2024
• Selected as one of 15 scholarship recipients (1% acceptance rate) for the Erasmus Mundus Joint Master in Artificial Intelligence.	
Dean’s Awards University of Ljubljana	2023 – 2024
• Received the Dean’s Award for Outstanding Academic Achievement, along with awards for exemplary academic performance in my second and third years of bachelor’s studies (top 2% of students).	

LICENSES & CERTIFICATIONS

Programming of GPUs - CUDA Issued by EuroCC SLING. Credential ID 202311-23	Jan. 2024
Fundamentals of Supercomputing Issued by EuroCC SLING. Credential ID 202303-4	Apr. 2023

TECHNICAL SKILLS & LANGUAGES

Programming: Python, Java, C/C++, SQL, Bash
Machine Learning: PyTorch, PyTorch Lightning, JAX, Hugging Face Transformers and Accelerate, scikit-learn
Infrastructure: Git, Docker, Kubernetes, Azure, Slurm, Neo4j, $\text{\LaTeX}$
Languages: English (IELTS 8.5/9.0), German (Elementary), Slovene (Advanced), Serbian (Native)